

Right Angle Triangle Pattern

```
In [3]: ▶ for i in range(1,6):
          for j in range(i):
              print("*",end=" ")
          print()
```

```
*
* *
* * *
* * * *
* * * * *
```

Inverted right angle triangle

```
In [5]: ▶ for i in range(1,6):
          for j in range(6-i):
              print("*",end=" ")
          print()
```

```
* * * * *
* * * *
* * *
* *
*
```

pyramid pattern

```
In [14]: ▶ for i in range(1,10):
           for j in range(i):
               if i%2==0:
                   continue
               print("*",end=" ")
           print()
```

```
*
* * *
* * * * *
* * * * * * *
* * * * * * * * *
```

```
In [15]: ▶ for i in range(1,6):
           print('*'(5-i)+'*'(2*i-1))
```

```
*
***
*****
*****
*****
```

```
In [18]: ▶ for i in range(1,10):
           if i%2==0:
               continue
           for j in range(i):
               print("*",end=" ")
           print()
```

```
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* * * * * * * * *
```

Inverted pyramid pattern

```
In [23]: ▶ for i in range(1,10):
           if i%2==0:
               continue
           for j in range(10-i):
               print("*",end=" ")
           print()
```

```
* * * * * * * * *
* * * * * * *
* * * * *
* * *
*
```

Diamond pattern

```
In [ ]: ▶ for i in range(1,10):
           if i%2==0:
               continue
           for j in range(10-i):
               print("*",end=" ")
           print()
```

```
In [5]: ► n=int(input("enter the diamond size"))

for i in range(1,n):
    if i%2==0:
        continue
    for j in range(i):
        print("*",end=" ")
    print()
for i in range(1,n):
    if i%2==0:
        continue
    for j in range(n-1-i):
        print("*",end=" ")
    print()
```

```
enter the diamond size10
*
* * *
* * * * *
* * * * * * *
* * * * * * * * *
* * * * * * * * *
* * * * * *
* * * *
* *
```

```
In [5]: ► n=int(input("Enter the diamond size"))
for i in range(1,n+1,2):
    print("***i")
for i in range(n-2,0,-2):
    print("***i")
```

```
Enter the diamond size9
*
***
*****
*****
*****
*****
***
*
```

Hallow square pattern

```
In [17]: ▶ for i in range(5):
            for j in range(5):
                if i==0 or i==4 or j==0 or j==4:
                    print("*",end=" ")
                else:
                    print(' ',end=" ")
            print()
```

```
*****
*     *
*     *
*     *
*****
```

```
In [19]: ▶ for i in range(5):
            for j in range(5):
                print("*",end=" ")
            print()
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
In [37]: ▶ n=int(input("enter a number"))
            for i in range(1,n):
                for j in range(1,n+1-i):
                    print(j,end=" ")
                print()
```

```
enter a number6
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
In [39]: ▶ n=int(input("enter a number"))
            for i in range(1,n):
                for j in range(1,i+1):
                    print(j,end=" ")
                print()
```

```
enter a number6
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
In [46]: ▶ n=int(input("enter a number"))

for i in range(1,n):
    for j in range(i,i*2):
        print(j,end=" ")
    print()
```

```
enter a number8
1
2 3
3 4 5
4 5 6 7
5 6 7 8 9
6 7 8 9 10 11
7 8 9 10 11 12 13
```

```
In [49]: ▶ num=1
for i in range(1,6):
    for j in range(1,i+1):
        print(num,end=' ')
        num+=1
    print()
```

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

```
In [53]: ▶ for i in range(5):
    for j in range(i+1):
        print("*",end="")
    else:
        print('',end=" ")
    print()
```

```
*
**
***
****
*****
```

```
In [58]: ▶ for i in range(1,6):
            for j in range(1,i+1):
                if j==1 or j==i or i==5:
                    print("*",end=" ")
                else:
                    print(" ",end=" ")
            print()
```

```
*
* *
*  *
*   *
* * * * *
```

```
In [62]: ▶ for i in range(1,6):
            for j in range(5-i):
                print(' ',end=' ')
            for j in range(2*i-1):
                if j==0 or j==2*i-2 or i==5:
                    print('*', end=' ')
                else:
                    print(' ', end=' ')
            print()
```

```
      *
    *  *
  *    *
*      *
* * * * * * * * *
```

```
In [68]: ► for i in range(1,6):
            for j in range(5-i):
                print(' ',end=' ')
            for j in range(2*i-1):
                if j==0 or j==2*i-2:
                    print('*', end=' ')
                else:
                    print(' ', end=' ')
            print()
        for i in range(4,0,-1):
            for j in range(5-i):
                print(' ',end=' ')
            for j in range(2*i-1):
                if j==0 or j==2*i-2:
                    print('*', end=' ')
                else:
                    print(' ', end=' ')
            print()
```

```
      *
    *   *
  *       *
*         *
*       *
*     *
*   *
* *
 *
```

```

In [70]: ▶ for i in range(1,6):
            for j in range(5-i):
                print(' ',end=' ')
            for j in range(2*i-1):
                if j==0 or j==2*i-2:
                    print(i, end=' ')
                else:
                    print(' ', end=' ')
            print()
        for i in range(4,0,-1):
            for j in range(5-i):
                print(' ',end=' ')
            for j in range(2*i-1):
                if j==0 or j==2*i-2:
                    print(i, end=' ')
                else:
                    print(' ', end=' ')
            print()

```

```

      1
    2 2
  3   3
4    4
5      5
  4    4
    3   3
      2 2
        1

```

```
In [74]: ▶ n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print(j,end=' ')
    for j in range(2*(n-i)):
        print(' ',end=" ")
    for j in range(1,i+1):
        print(j,end=' ')
    print()
for i in range(n,0,-1):
    for j in range(1,i+1):
        print(j,end=' ')
    for j in range(2*(n-i)):
        print(' ',end=" ")
    for j in range(1,i+1):
        print(j,end=' ')
    print()
```

```
1                1
1 2              1 2
1 2 3            1 2 3
1 2 3 4          1 2 3 4
1 2 3 4 5 1 2 3 4 5
1 2 3 4 5 1 2 3 4 5
1 2 3 4          1 2 3 4
1 2 3            1 2 3
1 2              1 2
1                1
```

```
In [75]: ▶ n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print('*',end=' ')
    for j in range(2*(n-i)):
        print(' ',end=" ")
    for j in range(1,i+1):
        print('*',end=' ')
    print()
for i in range(n,0,-1):
    for j in range(1,i+1):
        print('*',end=' ')
    for j in range(2*(n-i)):
        print(' ',end=" ")
    for j in range(1,i+1):
        print('*',end=' ')
    print()
```

```
*
* *
* * *
* * * *
* * * * *
* * * * *
* * * * *
* * * *
* * *
* *
*
```

```
In [76]: n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print(j,end=' ')
    for j in range(2*(n-i)):
        print(' ',end=" ")
    for j in range(1,i+1):
        print(j,end=' ')
    print()
for i in range(n,0,-1):
    for j in range(1,i+1):
        print(j,end=' ')
    for j in range(2*(n-i)):
        print(' ',end=" ")
    for j in range(1,i+1):
        print(j,end=' ')
    print()
for i in range(1,n+1):
    for j in range(1,i+1):
        print('*',end=' ')
    for j in range(2*(n-i)):
        print(' ',end=" ")
    for j in range(1,i+1):
        print('*',end=' ')
    print()
for i in range(n,0,-1):
    for j in range(1,i+1):
        print('*',end=' ')
    for j in range(2*(n-i)):
        print(' ',end=" ")
    for j in range(1,i+1):
        print('*',end=' ')
    print()
```

```
1                1
1 2              1 2
1 2 3            1 2 3
1 2 3 4          1 2 3 4
1 2 3 4 5 1 2 3 4 5
1 2 3 4 5 1 2 3 4 5
1 2 3 4          1 2 3 4
1 2 3            1 2 3
1 2              1 2
1                1
*                *
* *              * *
* * *            * * *
* * * *          * * * *
* * * * * * * * * *
* * * * * * * * * *
* * * *          * * * *
* * *            * * *
* *              * *
*
```

```
In [7]: ▶ n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(1,2*i):
        if j==1 or j==2*i-1 or i==5:
            print(i, end=" ")
        else:
            print(' ',end=" ")
    print()
```

```
      1
     2 2
    3   3
   4     4
  5 5 5 5 5 5 5 5 5
```

```
In [8]: ▶ n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(1,2*i):
        if j==1 or j==2*i-1 or i==5:
            print("*", end=" ")
        else:
            print(' ',end=" ")
    print()
```

```
      *
     * *
    *   *
   *     *
  *       *
 * * * * * * * * *
```

```
In [10]: ▶ n=5
for i in range(n,0,-1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(2*i-1):
        print("*", end=" ")
    print()
```

```
* * * * * * * * *
 * * * * * * *
  * * * * *
   * * *
    *
```

```
In [11]: ▶ n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print("*",end=" ")
    print()
```

```
*
* *
* * *
* * * *
* * * * *
```

```
In [13]: ▶ n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print(j,end=" ")
    print()
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
In [2]: ▶ n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(" ",end=" ")
    for j in range(1,i+1):
        print(j,end=" ")
    print()
```

```
      1
     1 2
    1 2 3
   1 2 3 4
  1 2 3 4 5
```

```
In [3]: ▶ n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(" ",end=" ")
    for j in range(1,i+1):
        print("*",end=" ")
    print()
```

```
      *
     * *
    * * *
   * * * *
  * * * * *
```

In []: ▶